

Delta Method and Method of Moments

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Objectives of this lecture

Transport a limit theorem through a function. Given an asymptotically normal $\hat{\theta}_n$, we ask whether $\phi(\hat{\theta}_n)$ is asymptotically normal, and with which variance. The delta method answers this for any ϕ differentiable at the point, and shows that only the derivative at this point matters.

Prove it from the calculus of lecture 2. The proof only relies on a first order Taylor expansion, whose remainder we control with the transfer theorem for $o_{\mathbb{P}}$ and $O_{\mathbb{P}}$.

Use it twice. We apply it to the empirical variance, which is not an average of independent variables and thus escapes a direct use of the central limit theorem, and to a transformation chosen to make the limiting variance constant.

Build a first general estimation method. The method of moments turns the model into a system of equations, and we obtain its asymptotic normality by applying the delta method to the inverse of a moment map.

Results imported from lecture 2

We use all of the following results without proving them again.

Prohorov. If $X_n \xrightarrow{d} X$, then (X_n) is uniformly tight, i.e. $X_n = O_{\mathbb{P}}(1)$.

Transfer of o and O . If $X_n \xrightarrow{\mathbb{P}} 0$, $R(0) = 0$ and $\|R(h)\| = o(\|h\|^q)$ as $h \rightarrow 0$, then $R(X_n) = o_{\mathbb{P}}(\|X_n\|^q)$. Furthermore, $o_{\mathbb{P}}(O_{\mathbb{P}}(1)) = o_{\mathbb{P}}(1)$.

Asymptotic equivalence. If $X_n \xrightarrow{d} X$ and $\|X_n - Y_n\| \xrightarrow{\mathbb{P}} 0$, then $Y_n \xrightarrow{d} X$.

Slutsky and continuous mapping. If $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{\mathbb{P}} c$, then $(X_n, Y_n) \xrightarrow{d} (X, c)$, and $g(X_n) \xrightarrow{d} g(X)$ for g continuous on a set of full limit mass.

The central limit theorem. For i.i.d. X_i with $\mathbb{E}_X[\|X\|^2] < \infty$, we have $\sqrt{n}(\bar{X}_n - \mathbb{E}_X[X]) \xrightarrow{d} \mathcal{N}(0, \text{Cov}_X(X))$.

Notation for this lecture

Parameter and observations. The parameter is $\theta \in \Theta \subset \mathbb{R}^p$, the observations $X_1, X_2, \dots$ are i.i.d. copies of a random vector X of $\mathbb{R}^d$, and $\hat{\theta}_n$ denotes an estimator built from $X_1, \dots, X_n$.

Derivatives. For $\phi : \mathbb{R}^p \rightarrow \mathbb{R}^m$ differentiable at θ , we denote by $J\phi(\theta)$ its $m \times p$ Jacobian matrix at θ . When $m = 1$, $\nabla\phi(\theta)$ denotes the gradient column vector and $H\phi(\theta)$ the Hessian.

Model. A statistical model is $\{\mathcal{L}_\theta; \theta \in \Theta\}$, each $\mathcal{L}_\theta$ a law on $\mathbb{R}^d$. We write $\mathbb{E}_{X \sim \mathcal{L}_\theta}$, abbreviated $\mathbb{E}_\theta$, for expectations computed as if the parameter were θ , which is the only abbreviation of this lecture.

Sample quantities. We denote $\bar{X}_n = \frac{1}{n} \sum_{i \leq n} X_i$, and any sample dependent object carries its index (e.g. $\hat{\theta}_n$, $\hat{\mu}_{2,n}$, e_n).

Part I. The delta method

The question

Setup. Given a sequence of estimators $(\hat{\theta}_n)$ of a parameter θ and a mapping ϕ , it is natural to estimate $\phi(\theta)$ by $\phi(\hat{\theta}_n)$.

Question. We ask whether the asymptotic properties of $\hat{\theta}_n - \theta$ are inherited by $\phi(\hat{\theta}_n) - \phi(\theta)$.

Consistency. Here the answer is immediate. If $\hat{\theta}_n \xrightarrow{\mathbb{P}} \theta$ and ϕ is continuous at θ , then $\phi(\hat{\theta}_n) \xrightarrow{\mathbb{P}} \phi(\theta)$ by continuous mapping, since the continuity set contains the constant θ .

Rate and limit law. Let us now suppose that

$$\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{d} \mathcal{N}(0, \Sigma).$$

In contrast, nothing above says at what rate $\phi(\hat{\theta}_n) - \phi(\theta)$ goes to zero, nor what its limit law is.

The linear case

If ϕ is linear, given by an $m \times p$ matrix M , then $\phi(\hat{\theta}_n) - \phi(\theta) = M(\hat{\theta}_n - \theta)$, and continuous mapping with the continuous map $x \mapsto Mx$ yields

$$\sqrt{n}(M\hat{\theta}_n - M\theta) = M\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{d} M\mathcal{N}(0, \Sigma) = \mathcal{N}(0, M\Sigma M^\top).$$

The last equality follows from the characteristic function computation of lecture 2, since for $Z \sim \mathcal{N}(0, \Sigma)$, we have $\phi_{MZ}(t) = \phi_Z(M^\top t) = \exp(-\frac{1}{2}t^\top M\Sigma M^\top t)$.

The intuition of the delta method. A function differentiable at θ is, near θ , roughly its linearisation. Hence, since $\hat{\theta}_n$ concentrates at θ , only the behaviour of ϕ at θ should matter, and $J\phi(\theta)$ should play the role of M .

The theorem

Theorem (delta method). Let $\theta \in \mathbb{R}^p$ and let $\phi : \mathbb{R}^p \rightarrow \mathbb{R}^m$ be differentiable at θ . Let $(\hat{\theta}_n)$ be a sequence of random vectors of $\mathbb{R}^p$, X a random vector of $\mathbb{R}^p$, and (r_n) real numbers with $r_n \rightarrow \infty$. If

$$r_n(\hat{\theta}_n - \theta) \xrightarrow{d} X, \quad \text{then} \quad r_n \left(\phi(\hat{\theta}_n) - \phi(\theta) \right) \xrightarrow{d} (J\phi(\theta))X$$

and, more precisely,

$$r_n \left(\phi(\hat{\theta}_n) - \phi(\theta) \right) - r_n(J\phi(\theta))(\hat{\theta}_n - \theta) \xrightarrow{\mathbb{P}} 0.$$

Remark. The second statement is the stronger one, since it says that the linearisation is exact up to a term negligible in probability at the scale r_n . Hence, the proof establishes the second statement and deduces the first from it.

Proof, consistency of the estimator

Claim. $\hat{\theta}_n - \theta = o_{\mathbb{P}}(1)$.

By Prohorov, $r_n(\hat{\theta}_n - \theta) = O_{\mathbb{P}}(1)$, so for $\epsilon > 0$ there is $C_\epsilon > 0$ with

$$\mathbb{P}\left(r_n \|\hat{\theta}_n - \theta\| \geq C_\epsilon\right) \leq \epsilon \quad \text{for every } n.$$

Let us fix $M > 0$. Since $r_n \rightarrow \infty$, we have $r_n M \geq C_\epsilon$ for n large, and for those n

$$\mathbb{P}\left(\|\hat{\theta}_n - \theta\| \geq M\right) = \mathbb{P}\left(r_n \|\hat{\theta}_n - \theta\| \geq r_n M\right) \leq \mathbb{P}\left(r_n \|\hat{\theta}_n - \theta\| \geq C_\epsilon\right) \leq \epsilon.$$

Hence, $\limsup_n \mathbb{P}(\|\hat{\theta}_n - \theta\| \geq M) \leq \epsilon$ for every $\epsilon > 0$, which proves the claim.

Shorter route. Since $\hat{\theta}_n - \theta = \frac{1}{r_n} r_n(\hat{\theta}_n - \theta)$ with $\frac{1}{r_n} \rightarrow 0$ deterministic, Slutsky yields convergence in law to $0 \cdot X = 0$, hence in probability since the limit is constant.

Proof, negligibility of the remainder

We consider the remainder of the first order expansion at θ ,

$$R(h) = \phi(\theta + h) - \phi(\theta) - (J\phi(\theta))h, \quad \text{so that} \quad R(0) = 0 \quad \text{and} \quad \|R(h)\| = o(\|h\|) \text{ as } h \rightarrow 0,$$

the second statement being the definition of differentiability at θ .

Since $\hat{\theta}_n - \theta \xrightarrow{\mathbb{P}} 0$, the transfer theorem with $q = 1$ yields $R(\hat{\theta}_n - \theta) = o_{\mathbb{P}}(\|\hat{\theta}_n - \theta\|)$, and multiplying by r_n , we obtain

$$r_n R(\hat{\theta}_n - \theta) = o_{\mathbb{P}}\left(r_n \|\hat{\theta}_n - \theta\|\right) = o_{\mathbb{P}}(O_{\mathbb{P}}(1)) = o_{\mathbb{P}}(1),$$

where we used Prohorov once more for $r_n(\hat{\theta}_n - \theta) = O_{\mathbb{P}}(1)$.

Eventually, since $r_n R(\hat{\theta}_n - \theta) = r_n(\phi(\hat{\theta}_n) - \phi(\theta)) - r_n(J\phi(\theta))(\hat{\theta}_n - \theta)$, we obtain the second statement of the theorem.

Proof, from the linearisation to the limit law

We denote $D_n = r_n(\phi(\hat{\theta}_n) - \phi(\theta)) - r_n(J\phi(\theta))(\hat{\theta}_n - \theta)$, so that $D_n \xrightarrow{\mathbb{P}} 0$ by the previous slide.

Since $r_n(\hat{\theta}_n - \theta) \xrightarrow{d} X$ and D_n converges in probability to the constant 0, Slutsky yields the joint convergence

$$\left(r_n(\hat{\theta}_n - \theta), D_n \right) \xrightarrow{d} (X, 0).$$

Since the map $g(x, y) = (J\phi(\theta))x + y$ is continuous on $\mathbb{R}^p \times \mathbb{R}^m$, continuous mapping yields

$$r_n \left(\phi(\hat{\theta}_n) - \phi(\theta) \right) = g \left(r_n(\hat{\theta}_n - \theta), D_n \right) \xrightarrow{d} g(X, 0) = (J\phi(\theta))X.$$

Remark. We can also conclude with asymptotic equivalence, since $r_n(J\phi(\theta))(\hat{\theta}_n - \theta) \xrightarrow{d} (J\phi(\theta))X$ by continuous mapping and the two sequences differ by $D_n \xrightarrow{\mathbb{P}} 0$.

A mnemonic for the delta method

Informally, and as a mnemonic rather than an argument, we have

$$r_n \left(\phi(\hat{\theta}_n) - \phi(\theta) \right) \simeq \underbrace{\left(r_n(\hat{\theta}_n - \theta) \right)}_{\simeq X} \times \underbrace{\left(\frac{\phi(\hat{\theta}_n) - \phi(\theta)}{\hat{\theta}_n - \theta} \right)}_{\simeq J\phi(\theta) \text{ for } n \text{ large}}.$$

The first factor is the assumption, while the second is the difference quotient, which tends to the derivative since $\hat{\theta}_n$ tends to θ . The theorem states that this splitting is legitimate, and its proof checks that the error in the second factor is small enough to survive multiplication by r_n .

Reading the hypotheses

Differentiability is needed at one point only. We assume nothing about ϕ away from θ , not even continuity, which is possible since $\hat{\theta}_n$ concentrates at θ .

The limit X need not be gaussian. The theorem holds for an arbitrary limit in law. However, in every application below, X is gaussian since it comes from the central limit theorem, and then $(J\phi(\theta))X \sim \mathcal{N}(0, J\phi(\theta)\Sigma(J\phi(\theta))^\top)$.

The rate is unchanged. Since the same r_n appears on both sides, $\phi(\hat{\theta}_n)$ converges at the rate of $\hat{\theta}_n$, with a different constant.

The case of a vanishing Jacobian. If $J\phi(\theta) = 0$, the limit is the constant 0 and the theorem, while true, only says that the rate r_n is too slow. We give the right rate in Part III.

Part II. Estimating a variance

The problem

Let $X_1, X_2, \dots$ be i.i.d. copies of a real X with $\mathbb{E}_X[|X|^4] < \infty$. We denote $\mu_1 = \mathbb{E}_X[X]$ and, for $\ell \in \{2, 3, 4\}$, the centered moments $\mu_\ell = \mathbb{E}_X[(X - \mu_1)^\ell]$.

We estimate the mean by $\hat{\mu}_{1,n} = \bar{X}_n$ and the variance μ_2 by the empirical variance

$$\hat{\mu}_{2,n} = \frac{1}{n} \sum_{i \leq n} (X_i - \bar{X}_n)^2.$$

The difficulty. Since $\hat{\mu}_{1,n}$ is an average of i.i.d. variables, the central limit theorem applies to it directly. In contrast, $\hat{\mu}_{2,n}$ is not such an average, since its summands all involve $\bar{X}_n$, hence all the data, and they are neither independent nor identically distributed as written.

An identity for the empirical variance

By expanding the square and then using that a variance is unchanged by a translation, we get

$$\hat{\mu}_{2,n} = \frac{1}{n} \sum_{i \leq n} X_i^2 - \left(\frac{1}{n} \sum_{i \leq n} X_i \right)^2 = \frac{1}{n} \sum_{i \leq n} (X_i - \mu_1)^2 - \left(\frac{1}{n} \sum_{i \leq n} (X_i - \mu_1) \right)^2.$$

The right side is thus a fixed function of two averages of i.i.d. variables. Let us set

$$Y_i = \begin{pmatrix} X_i - \mu_1 \\ (X_i - \mu_1)^2 \end{pmatrix} \in \mathbb{R}^2, \quad \phi(x, y) = y - x^2,$$

so that $\hat{\mu}_{2,n} = \phi(\bar{Y}_n)$ exactly, and $\mu_2 = \phi(0, \mu_2)$.

The central limit theorem for the pair

Since $\mathbb{E}_X[(X - \mu_1)] = 0$ and $\mathbb{E}_X[(X - \mu_1)^2] = \mu_2$, the vector Y_1 has mean $(0, \mu_2)^\top$ and covariance

$$\text{Cov}_X(Y_1) = \begin{pmatrix} \mu_2 & \mu_3 \\ \mu_3 & \mu_4 - \mu_2^2 \end{pmatrix} =: \Sigma_Y,$$

which is finite since $\mathbb{E}_X[|X|^4] < \infty$. The entries are $\text{Var}_X(X - \mu_1) = \mu_2$, then $\mathbb{E}_X[(X - \mu_1)^3] = 0$, $\mu_2 = \mu_3$, and $\text{Var}_X((X - \mu_1)^2) = \mu_4 - \mu_2^2$.

Since the Y_i are i.i.d. and square integrable, the central limit theorem in $\mathbb{R}^2$ yields

$$\sqrt{n} \left(\bar{Y}_n - \begin{pmatrix} 0 \\ \mu_2 \end{pmatrix} \right) \xrightarrow{d} \mathcal{N}(0, \Sigma_Y).$$

Applying the delta method

The map $\phi(x, y) = y - x^2$ is differentiable with $J\phi(x, y) = (-2x \ 1)$, so that at the evaluation point, $J\phi(0, \mu_2) = (0 \ 1)$. The delta method with $r_n = \sqrt{n}$ then yields

$$\sqrt{n}(\hat{\mu}_{2,n} - \mu_2) = \sqrt{n}(\phi(\bar{Y}_n) - \phi(0, \mu_2)) \xrightarrow{d} \mathcal{N}\left(0, (0 \ 1)\Sigma_Y \begin{pmatrix} 0 \\ 1 \end{pmatrix}\right),$$

and the quadratic form picks out the lower right entry:

$$\sqrt{n}(\hat{\mu}_{2,n} - \mu_2) \xrightarrow{d} \mathcal{N}(0, \mu_4 - \mu_2^2).$$

The first coordinate of Y has disappeared since $J\phi$ vanishes there. In other words, the fluctuation of $\bar{X}_n$ contributes nothing at first order to the fluctuation of the empirical variance.

Remarks on the result

The limiting variance is $\text{Var}_X((X - \mu_1)^2)$. It is zero if and only if $(X - \mu_1)^2$ is almost surely constant, i.e. when X takes only the two values $\mu_1 \pm \sqrt{\mu_2}$, and for every other law, the limit is nondegenerate.

In the gaussian case, $\mu_4 = 3\mu_2^2$, so the limit is $\mathcal{N}(0, 2\mu_2^2)$, and the relative precision $\sqrt{n}(\hat{\mu}_{2,n} - \mu_2)/\mu_2$ has limiting variance 2 whatever the scale.

The bias correction. With $s_n^2 = \frac{n}{n-1}\hat{\mu}_{2,n}$ the unbiased version, we have $\sqrt{n}(s_n^2 - \hat{\mu}_{2,n}) = \frac{\sqrt{n}}{n-1}\hat{\mu}_{2,n} \xrightarrow{\mathbb{P}} 0$, so that by asymptotic equivalence, s_n^2 has the same limit law. Hence, unbiasedness at fixed n has no asymptotic content here.

Part III. Two refinements

The case of a vanishing Jacobian

If $J\phi(\theta) = 0$, the delta method yields $r_n(\phi(\hat{\theta}_n) - \phi(\theta)) \xrightarrow{d} 0$, a true but empty statement which only says that the rate r_n is too slow. We now show that the correct rate is r_n^2 .

Theorem (second order delta method). Let $\phi : \mathbb{R}^p \rightarrow \mathbb{R}$ be twice differentiable at θ with $\nabla\phi(\theta) = 0$, and let $r_n(\hat{\theta}_n - \theta) \xrightarrow{d} X$ with $r_n \rightarrow \infty$. Then

$$r_n^2 \left(\phi(\hat{\theta}_n) - \phi(\theta) \right) \xrightarrow{d} \frac{1}{2} X^\top (H\phi(\theta)) X.$$

The limit is no longer gaussian. Even for X gaussian, a quadratic form in a gaussian vector is a weighted sum of independent χ_1^2 variables, with weights the eigenvalues of $\frac{1}{2}\Sigma^{1/2}(H\phi(\theta))\Sigma^{1/2}$.

Proof of the second order version

Using $\nabla\phi(\theta) = 0$, we consider the second order remainder at θ ,

$$R(h) = \phi(\theta + h) - \phi(\theta) - \frac{1}{2}h^\top (H\phi(\theta))h, \quad R(0) = 0, \quad |R(h)| = o(\|h\|^2) \text{ as } h \rightarrow 0,$$

the last being the Taylor expansion at order two of a function twice differentiable at θ .

Since $\hat{\theta}_n - \theta \xrightarrow{\mathbb{P}} 0$, the transfer theorem with $q = 2$ yields $R(\hat{\theta}_n - \theta) = o_{\mathbb{P}}(\|\hat{\theta}_n - \theta\|^2)$, hence

$$r_n^2 R(\hat{\theta}_n - \theta) = o_{\mathbb{P}}\left(\|r_n(\hat{\theta}_n - \theta)\|^2\right) = o_{\mathbb{P}}(O_{\mathbb{P}}(1)) = o_{\mathbb{P}}(1),$$

since $r_n(\hat{\theta}_n - \theta) = O_{\mathbb{P}}(1)$ by Prohorov and a continuous image of a tight sequence is tight.

Eventually, $r_n^2(\phi(\hat{\theta}_n) - \phi(\theta)) = \frac{1}{2}(r_n(\hat{\theta}_n - \theta))^\top (H\phi(\theta))(r_n(\hat{\theta}_n - \theta)) + o_{\mathbb{P}}(1)$, and continuous mapping with $x \mapsto \frac{1}{2}x^\top (H\phi(\theta))x$ together with asymptotic equivalence concludes the proof.

Example, a proportion at one half

Let $X_1, \dots, X_n$ be i.i.d. $\text{Ber}(p)$ and $\phi(p) = p(1 - p)$, the variance of a single observation. Then $\phi'(p) = 1 - 2p$, which vanishes only at $p = 1/2$.

Away from one half, the delta method yields $\sqrt{n}(\bar{X}_n(1 - \bar{X}_n) - p(1 - p)) \xrightarrow{d} \mathcal{N}(0, (1 - 2p)^2 p(1 - p))$.

At $p = 1/2$, the first order limit is degenerate, and applying the second order version with $\phi'' = -2$ and $\sqrt{n}(\bar{X}_n - \frac{1}{2}) \xrightarrow{d} \mathcal{N}(0, \frac{1}{4})$, we obtain

$$n(\bar{X}_n(1 - \bar{X}_n) - \frac{1}{4}) \xrightarrow{d} -X^2 \stackrel{d}{=} -\frac{1}{4}\chi_1^2, \quad X \sim \mathcal{N}(0, \frac{1}{4}).$$

The rate improves from $n^{-1/2}$ to n^{-1} and the limit is one sided, as expected since $\bar{X}_n(1 - \bar{X}_n) \leq 1/4$ always, so that the error can only have one sign.

Variance stabilisation

Let us take $p = m = 1$ and suppose that $\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{d} \mathcal{N}(0, \sigma^2(\theta))$, with a limiting variance depending on the unknown θ . For ϕ differentiable, the delta method yields

$$\sqrt{n} \left(\phi(\hat{\theta}_n) - \phi(\theta) \right) \xrightarrow{d} \mathcal{N} \left(0, \phi'(\theta)^2 \sigma^2(\theta) \right).$$

The recipe. We choose ϕ with $\phi'(\theta) = 1/\sigma(\theta)$ for every θ , i.e. a primitive of $1/\sigma$. Then the limit is $\mathcal{N}(0, 1)$, the same for every θ , so that the unknown no longer appears in the limit law.

Advantages. Since no plug-in estimate of the variance is needed, the studentisation step of lecture 2 disappears. Furthermore, the normal approximation is usually accurate at smaller n , since the transformed statistic is closer to symmetric.

The arcsine transformation

For $X_1, \dots, X_n$ i.i.d. $\text{Ber}(p)$ with $p \in (0, 1)$, $\sigma^2(p) = p(1 - p)$, so that the recipe asks for a primitive of $1/\sqrt{p(1 - p)}$, namely

$$\phi(p) = 2 \arcsin \sqrt{p}, \quad \phi'(p) = \frac{2}{\sqrt{1 - p}} \cdot \frac{1}{2\sqrt{p}} = \frac{1}{\sqrt{p(1 - p)}},$$

and hence $\sqrt{n} \left(2 \arcsin \sqrt{\bar{X}_n} - 2 \arcsin \sqrt{p} \right) \xrightarrow{d} \mathcal{N}(0, 1)$.

The interval. With r_α the quantile of order $1 - \alpha/2$ of $\mathcal{N}(0, 1)$, inverting ϕ , which is increasing from $[0, 1]$ onto $[0, \pi]$, yields an asymptotic confidence interval of level $1 - \alpha$ for p :

$$\left[\sin^2 \left(\arcsin \sqrt{\bar{X}_n} - \frac{r_\alpha}{2\sqrt{n}} \right), \sin^2 \left(\arcsin \sqrt{\bar{X}_n} + \frac{r_\alpha}{2\sqrt{n}} \right) \right] \cap [0, 1].$$

Other classical cases. Poisson with $\sigma^2(\lambda) = \lambda$ leads to $\phi(\lambda) = 2\sqrt{\lambda}$, and the correlation coefficient to the Fisher transformation $\phi(\rho) = \text{arctanh}(\rho)$.

Part IV. Statistical models

The model

Definition. A parametric statistical model is a family $\{\mathcal{L}_\theta; \theta \in \Theta\}$ with $\Theta \subset \mathbb{R}^p$, each $\mathcal{L}_\theta$ a law on $\mathbb{R}^d$. In other words, it is a set of candidate laws for X_1 .

Well specified. We assume throughout that the model contains the truth, i.e. that there is θ_0 in the interior of Θ such that $X_1, \dots, X_n$ are i.i.d. of law $\mathcal{L}_{\theta_0}$, and our goal is to estimate θ_0 .

Notation. $\mathbb{E}_\theta, \mathbb{P}_\theta, \text{Cov}_\theta$ denote the expectation, probability and covariance computed as if the parameter were θ . For instance, with $d = 1, \Theta = (0, \infty)$ and $\mathcal{L}_\theta = \mathcal{N}(0, \theta)$, we have

$$\mathbb{E}_3[X_1^2] = \int_{\mathbb{R}} x^2 d\mathcal{L}_3(x) = \mathbb{E}_{Z \sim \mathcal{N}(0,3)}[Z^2] = 3.$$

We keep writing $\mathbb{E}_{\theta_0} = \mathbb{E}_X$, since $\mathcal{L}_{\theta_0}$ really is the law of the data.

Identifiability

A method that matches p features of the law can only work if those features separate the parameters. Given $f_1, \dots, f_p : \mathbb{R}^d \rightarrow \mathbb{R}$, we denote

$$e(\theta) = \begin{pmatrix} \mathbb{E}_\theta[f_1(X_1)] \\ \vdots \\ \mathbb{E}_\theta[f_p(X_1)] \end{pmatrix} \in \mathbb{R}^p.$$

Identifiability from the moments is the requirement that $e(\theta) \neq e(\theta_0)$ for every $\theta \neq \theta_0$. Without it, two parameters produce identical moment equations, and no sample size separates them.

The theorem below assumes more, namely that e is continuously differentiable near θ_0 with $Je(\theta_0)$ invertible. By the inverse function theorem, this forces local injectivity and makes e locally invertible, which is what the estimator relies on.

Number of moment functions. There are as many moment functions as parameters (p of each), since $Je(\theta_0)$ is $p \times p$ and has to be invertible. With fewer, the system is underdetermined, whereas with more, it is generally incompatible.

Part V. The method of moments

The method

Method of moments. We choose $f_1, \dots, f_p : \mathbb{R}^d \rightarrow \mathbb{R}$ and we look for a parameter $\hat{\theta}$ at which the theoretical moments equal the empirical ones:

$$\left\{ \begin{array}{l} \frac{1}{n} \sum_{i \leq n} f_1(X_i) = \mathbb{E}_{\hat{\theta}}[f_1(X_1)] \\ \vdots \\ \frac{1}{n} \sum_{i \leq n} f_p(X_i) = \mathbb{E}_{\hat{\theta}}[f_p(X_1)] \end{array} \right.$$

Compactly. With e as on the previous slide and $e_n = \left(\frac{1}{n} \sum_{i \leq n} f_1(X_i), \dots, \frac{1}{n} \sum_{i \leq n} f_p(X_i) \right)^\top$, the system takes the form $e(\hat{\theta}) = e_n$, so that $\hat{\theta} = e^{-1}(e_n)$ whenever the inverse is available.

Intuition. The law of large numbers yields $e_n \xrightarrow{\mathbb{P}} e(\theta_0)$, and since e^{-1} is continuous near $e(\theta_0)$, the solution should be close to θ_0 . The analysis below makes this precise and adds the rate.

Example, the gaussian model

Exercise. Let $\Theta = \mathbb{R} \times (0, \infty)$, $\theta = (m, \sigma^2)$ and $\mathcal{L}_\theta = \mathcal{N}(m, \sigma^2)$. With $f_1(x) = x$ and $f_2(x) = x^2$, find the estimator.

Solution. We have $\mathbb{E}_\theta[f_1(X_1)] = m$ and $\mathbb{E}_\theta[f_2(X_1)] = m^2 + \sigma^2$, so that the system is

$$\frac{1}{n} \sum_{i \leq n} X_i = \hat{m}_n \quad \text{and} \quad \frac{1}{n} \sum_{i \leq n} X_i^2 = \hat{m}_n^2 + \hat{\sigma}_n^2.$$

It has the unique solution

$$\hat{m}_n = \bar{X}_n \quad \text{and} \quad \hat{\sigma}_n^2 = \frac{1}{n} \sum_{i \leq n} X_i^2 - \bar{X}_n^2 = \frac{1}{n} \sum_{i \leq n} (X_i - \bar{X}_n)^2,$$

i.e. the empirical mean and the empirical variance. Their joint asymptotic normality follows from Part II, and the theorem below recovers it from the general statement.

Asymptotic normality

Theorem. With e as above, assume

1. θ_0 is in the interior of Θ and there is $\epsilon > 0$ with $B(\theta_0, \epsilon) \subset \Theta$,
2. e is continuously differentiable on $B(\theta_0, \epsilon)$ with $Je(\theta_0)$ invertible,
3. $\mathbb{E}_X[|f_j(X_1)|^2] < \infty$ for $j = 1, \dots, p$.

Then we can define a random vector $\hat{\theta}_n$ that solves the moment system with probability going to 1, and

$$\sqrt{n} \left(\hat{\theta}_n - \theta_0 \right) \xrightarrow{d} \mathcal{N} \left(0, (Je(\theta_0))^{-1} \Sigma_f (Je(\theta_0))^{-\top} \right),$$

where Σ_f is the covariance matrix of the random vector $(f_1(X_1), \dots, f_p(X_1))$.

Proof, local inversion

Since e is continuously differentiable near θ_0 with $Je(\theta_0)$ invertible, the inverse function theorem yields open sets $U \ni \theta_0$ and $V \ni e(\theta_0)$ such that

$e : U \rightarrow V$ is a C^1 diffeomorphism, and $(Je^{-1})(v) = (Je(u))^{-1}$ for $v = e(u) \in V$.

In particular, at the point of interest,

$$(Je^{-1})(e(\theta_0)) = (Je(\theta_0))^{-1}.$$

We only use hypotheses 1 and 2 here, and they allow us to apply the delta method to $\phi = e^{-1}$ at $e(\theta_0)$.

Proof, existence with probability going to one

By hypothesis 3, each $f_j(X_i)$ is square integrable, hence integrable on a probability space, so that the law of large numbers applies coordinatewise:

$$e_n \xrightarrow{\mathbb{P}} e(\theta_0).$$

Since V is open and contains $e(\theta_0)$, this yields $\mathbb{P}(e_n \in V) \rightarrow 1$, and we define

$$\hat{\theta}_n = \begin{cases} e^{-1}(e_n) & \text{if } e_n \in V \\ \text{an arbitrary value} & \text{if } e_n \notin V. \end{cases}$$

On the event $\{e_n \in V\}$, we have $e(\hat{\theta}_n) = e_n$, so that $\hat{\theta}_n$ solves the moment system, and this event has probability going to 1. The estimator is to be understood in this sense, since the system may have no solution for small n , which does not matter asymptotically.

Proof, removing the arbitrary value

We define a companion sequence that always lies in V ,

$$\tilde{e}_n = \begin{cases} e_n & \text{if } e_n \in V \\ e(\theta_0) & \text{if } e_n \notin V, \end{cases}$$

so that $e^{-1}(\tilde{e}_n)$ is always defined, and equals $\hat{\theta}_n$ on $\{e_n \in V\}$.

Since $\theta_0 = e^{-1}(e(\theta_0))$, the two sequences $\sqrt{n}(\hat{\theta}_n - \theta_0)$ and $\sqrt{n}(e^{-1}(\tilde{e}_n) - e^{-1}(e(\theta_0)))$ coincide on $\{e_n \in V\}$, hence for every $\epsilon > 0$

$$\mathbb{P} \left(\left\| \sqrt{n}(\hat{\theta}_n - \theta_0) - \sqrt{n}(e^{-1}(\tilde{e}_n) - e^{-1}(e(\theta_0))) \right\| \geq \epsilon \right) \leq \mathbb{P}(e_n \notin V) \rightarrow 0.$$

By asymptotic equivalence, it is thus enough to find the limit law of $\sqrt{n}(e^{-1}(\tilde{e}_n) - e^{-1}(e(\theta_0)))$.

Proof, the delta method

Since the vectors $(f_1(X_i), \dots, f_p(X_i)), i \leq n$, are i.i.d., square integrable by hypothesis 3, with mean $e(\theta_0)$ and covariance Σ_f , the central limit theorem yields

$$\sqrt{n}(e_n - e(\theta_0)) \xrightarrow{d} \mathcal{N}(0, \Sigma_f),$$

and since $\tilde{e}_n$ differs from e_n only on an event of vanishing probability, asymptotic equivalence yields the same limit for $\sqrt{n}(\tilde{e}_n - e(\theta_0))$.

We now apply the delta method to $\phi = e^{-1}$, differentiable at $e(\theta_0)$, with $r_n = \sqrt{n}$:

$$\sqrt{n}(e^{-1}(\tilde{e}_n) - e^{-1}(e(\theta_0))) \xrightarrow{d} (Je(\theta_0))^{-1} \mathcal{N}(0, \Sigma_f) = \mathcal{N}(0, (Je(\theta_0))^{-1} \Sigma_f (Je(\theta_0))^{-\top}).$$

Together with the previous slide, this concludes the proof.

Reading the asymptotic variance

Let us take $p = 1$, so that the asymptotic variance is the number

$$\frac{\text{Var}_X(f_1(X_1))}{(e'(\theta_0))^2}, \quad e(\theta) = \mathbb{E}_\theta[f_1(X_1)].$$

The numerator is the noise. The smaller the fluctuation of $f_1(X_1)$, the more precisely the empirical moment estimates the theoretical one.

The denominator is the signal. The larger $e'(\theta_0)$, the more the theoretical moment moves when the parameter moves, and thus the better f_1 discriminates between θ_0 and its neighbours.

A good moment function thus has an expectation that varies fast with θ and a small variance under $\mathcal{L}_{\theta_0}$. However, the method does not prescribe f_1 , and different choices lead to different variances, none of them optimal in general.

The limits of the method

Example. Let $\Theta = (0, \infty)$, $\mathcal{L}_\theta = \text{Unif}([0, \theta])$ and $f_1(x) = x$. Then $e(\theta) = \theta/2$, so that $e'(\theta_0) = 1/2$ is invertible, $\Sigma_f = \text{Var}_X(X) = \theta_0^2/12$, and the theorem applies:

$$\hat{\theta}_n = 2\bar{X}_n, \quad \sqrt{n} \left(\hat{\theta}_n - \theta_0 \right) \xrightarrow{d} \mathcal{N} \left(0, \frac{\theta_0^2}{3} \right).$$

Comparison with the maximum of the sample. For $M_n = \max_{i \leq n} X_i$, a direct computation yields $\mathbb{P}(n(\theta_0 - M_n) > t) = (1 - t/(n\theta_0))^n \rightarrow e^{-t/\theta_0}$, so that $n(\theta_0 - M_n)$ converges in law to an exponential. Hence, the rate is $1/n$ rather than $1/\sqrt{n}$.

Interpretation. The method of moments is consistent and asymptotically normal here, but badly suboptimal, since the information about θ_0 lies in the largest observation rather than in an average. No choice of moment functions repairs this.

Summary of the lecture

One theorem. A limit in law passes through a function differentiable at the limit point, the derivative acting as a matrix. In other words, $r_n(\hat{\theta}_n - \theta) \xrightarrow{d} X$ implies $r_n(\phi(\hat{\theta}_n) - \phi(\theta)) \xrightarrow{d} (J\phi(\theta))X$, so that the rate is unchanged and only $J\phi(\theta)$ survives.

One proof, reusable. We Taylor expand, control the remainder with the transfer theorem for $\mathcal{O}_{\mathbb{P}}$, and conclude with Slutsky and continuous mapping. Changing $q = 1$ to $q = 2$ then yields the second order version for free.

Two uses. Reaching statistics that are not averages, such as the empirical variance, and choosing a transformation that removes the unknown from the limiting variance.

One method. The method of moments is $\hat{\theta}_n = e^{-1}(e_n)$, and its asymptotic normality follows from the central limit theorem for e_n , pushed through e^{-1} by the delta method. Its variance can be seen as noise over signal, and the method is not optimal.

Next. M and Z estimators, defined by maximising or by solving an equation rather than by inverting a map, and for which consistency itself has to be proved.

References

Van der Vaart, *Asymptotic Statistics*, Cambridge University Press, chapter 3 for the delta method and chapter 4 for moment estimators.

Bachoc, *Lecture notes, Asymptotic statistics*, University Paul Sabatier, sections 2 and 3, the course notes these slides follow.

Serfling, *Approximation Theorems of Mathematical Statistics*, chapter 3, for the delta method, variance stabilisation and the second order version.

Lehmann and Casella, *Theory of Point Estimation*, chapter 5, for the comparison between moment estimators and likelihood based ones.